

COM410 Programming in Practice

B1. Algorithm Performance

CHALLENGE

Prove the time complexity of the LinkedList add and remove operations

- Add a new class called **ListComplexityTest** to the **Analysis** project
- The **main()** method of **ListComplexityTest** should create a new **LinkedList** of **Integer** objects and generate **N** random integers $< N*10$, storing each in the **LinkedList** as it is generated.
- For each integer generated, measure the time taken in nanoseconds to store the element in the **LinkedList** and obtain the total time taken to add the elements.
- Now generate a further **N** random integers in the same range and attempt to remove an element of that value from the **LinkedList**, again calculating the total time to remove.
- Repeat for values of **N** of 10, 100, 1000, 10000 and 100000 elements and report for each value of **N**, the total time to add, the average time to add, the total time to remove and the average time to remove.

Please upload a screenshot of your solution running to the B1 Challenge submission link on Blackboard, by the deadline shown there.